A Literature Review of Defeasible Reasoning

Samukelisiwe Zwane
Department of Computer Science, University of Cape Town
Private Bag X3, Rondebosch, 7701, South Africa
zwnsam011@myuct.ac.za

Abstract

Defeasible reasoning, a type of non-monotonic reasoning that allows for the revision of conclusions when new or contradictory information emerges, reflecting the natural way humans update their beliefs. The KLM framework formalizes this behaviour through defeasible conditionals and ranked interpretations. Building on this foundation, Rational Closure offers the most conservative form of defeasible entailment, while Lexicographic Closure and Relevant Closure tackle its known limitation, by using more presumptive approaches. Despite the increasing importance of defeasible reasoning in artificial intelligence, there is no dedicated pedagogical tool to aid users in understanding how these algorithms derive conclusions. This review explores the theoretical foundations of defeasible reasoning, from Rational Closure to Lexicographic Closure and analyses existing logical and defeasible reasoning tools. The aim is to identify the gap that drives the development of a pedagogical defeasible reasoning tool. The focus of this literature review will be on Lexicographic Closure.

Keywords

Propositional logic, defeasible reasoning, entailment, Lexicographic Closure, non-monotonicity, Artificial Intelligence

1 Introduction

Humans and machines differ in the way in which they process information and derive conclusions based on their domain knowledge. Given new information it is intuitive for humans to come up with refined conclusions, however it is not as simple for machines. Knowledge Representation and Reasoning is a subfield of Artificial Intelligence that focuses on using symbolic representations to model different environments and derive conclusions or make decisions based on known premises [3]. Computers manage, sort and retrieve information from their knowledge bases (A set of real-world information) and are known to make decisions based on this. However, the complexities of the real world may contradict the strict adherence to deductive validity of inference drawn by these classical systems. Defeasible reasoning solves this by allowing for rational but not necessarily deductively valid conclusions, loosening the set structure of classical reasoning. A conclusion is considered defeasible when it is supported by available evidence but remains open to revision if new, contradictory information arises [18] This project aims to develop a pedagogical defeasible reasoning tool to help users understand the paths taken by agents to reach a particular conclusion.The tool will rely on theoretical frameworks for non-monotonic reasoning and practical implementations of defeasible reasoners. This review focuses on the Lexicographic Closure method for default reasoning. The review is structured as follows: Section 2 introduces propositional logic, covering syntax, semantics, and the limitations of classical monotonic reasoning. Section 3 discusses defeasible reasoning, the KLM framework, ranked interpretations, and the three main entailment methods: Rational Closure, Lexicographic Closure, and Relevant Closure. Section 4 examines existing pedagogical tools for classical logic and current defeasible reasoners, identifying the gap that motivates this work.

2 Propositional logic Background

Before introducing Defeasible Reasoning, we must establish the foundation upon which it was built, Propositional logic. Propositional logic is a mathematical formalization of natural language where statements are make up of words or other statements that cannot be broken down into simpler parts, called propositions. Each statement is assigned a value of True (T) or False (F) [8] and the construction of larger or more complex statements is done by using Boolean connective. [2] The truth of a complex statement depends solely on the truth values assigned to its atomic components and the meaning of connectives.[13] While propositional logic offers a precise and compact framework for representing and evaluating logical relationships, its monotonic nature means that once a conclusion is derived, it cannot be retracted. We first establish the syntax and semantics of propositional logic, as these form the foundation on which defeasible reasoning models are built.

2.1 Syntax

Propositional logic syntax is used to describe what statements are grammatically valid. The language makes use of atomic propositions and connectives to build propositional formulas. Propositional formulas are typically denoted by Greek letters α, β, γ and are defined over atoms using logical connectives, namely:

- conjunction (logical "and", $\wedge$)
- disjunction (logical "or", $\vee$)
- negation (logical "not", $\neg$)

Formally, the set propositional formulas is recursively defined such that every atom is a formula, the negation $\neg\alpha$ of any formula α is itself a formula and the conjunction $(\alpha \wedge \beta)$ and disjunction $(\alpha \vee \beta)$ of any two formulas are also formulas [3].

Additional connectives such as implication $(\rightarrow)$ and equivalence $(\leftrightarrow)$ are typically treated as abbreviations. Specifically, $\alpha \rightarrow \beta$ can also be defined as $\neg\alpha \vee \beta$. This formula construction becomes necessary when defeasible implications are translated using classical logic.

2.2 Semantics

Semantics informs whether a formula is true or false under a given interpretation (valuation) [2]. This valuation u is a function such that $u : P \rightarrow \{T, F\}$ [13], meaning it assigns True or False for every atom in the language. For the set P of propositions a and b,

$P = \{a, b\}$, the possible valuations $u \in \mathcal{U}$ can be represented using Boolean operators in a truth table as follows:

a	b	$a \vee b$	$a \wedge b$	$a \rightarrow b$	$a \leftrightarrow b$	$\neg a$
T	T	T	T	T	T	F
T	F	T	F	F	F	F
F	T	T	F	T	F	T
F	F	F	F	T	T	T

Table 1: A Truth table for Boolean operators

If an atom is true in a valuation, we can say that the valuation satisfies the atom[13]. Satisfaction is the foundation for defining logical consequence and it indicates that a statement is true under a specific valuation. A valuation u is said to satisfy a formula α, written $u \Vdash \alpha$, if α evaluates to true under the truth values that u assigns to its atoms.[1] For instance, given the formula $a \wedge b$, a valuation that assigns $u(a) = T$ and $u(b) = T$ satisfies it, whereas a valuation that assigns $u(a) = T$ and $u(b) = F$ does not.

Entailment extends this concept and can be defined as a relation between a pair of statements such that the truth of the second sentence follows from the truth of the first. A formula α is entailed by a set of formulas S, written $S \models \alpha$, if every valuation that satisfies all formulas in S also satisfies α [13]. In other words, there is no valuation that makes all formulas in S true while making α false. A Knowledge base $\mathcal{K}$ is a finite set of propositional formulas [13]. It gathers multiple pieces of information about a domain and allows us to reason about what they collectively imply.

2.3 Limitations of Classical Logic

Classical logic is monotonic, if $\mathcal{K} \models \alpha$, then $\mathcal{K} \cup \mathcal{K}' \models \alpha$ for any set of formulas $\mathcal{K}'$ [19]. This means that any conclusion α derived from a knowledge base K remains valid even when new information $\mathcal{K}'$ is added. New knowledge can never invalidate existing conclusions. While this ensures inference stability, it falls short in scenarios involving incomplete knowledge, where new information often contradicts previous conclusions.

This limitation is evident in the following example. Consider the following. Consider

$$\text{knowledge base } \mathcal{K} = \{c \rightarrow p, e \rightarrow c, e \rightarrow \neg p\}$$

Where c = "car", p = "uses petrol" and e = "electric car". From $e \rightarrow c$ and $c \rightarrow p$, The classical inference rules conclude $e \rightarrow p$. However, our knowledge base K also states $e \rightarrow \neg p$, which directly contradicts this conclusion. Classical logic therefore concludes that electric cars cannot exist, when that is not the case.

The real world is inherently complex, with exceptions and evolving circumstances that classical logic cannot adequately model. Therefore, any logic intended to formalize default reasoning must be non-monotonic, allowing for the revision of beliefs when the default assumptions they are based on are invalidated [19]

3 Defeasible reasoning

Defeasible Reasoning is characterized by conclusions that are supported by available evidence but remain open to revision in light of new, potentially contradictory information [18].Unlike classical reasoning, defeasible reasoning treats conclusions as conditional, holding them by default unless they are invalidated by more precise or specific information. For instance, without creating a contradiction, a defeasible reasoner would conclude that electric cars are an exception to the general default that "cars use petrol." The logic that aims to formalize this kind of reasoning is non-monotonic, allowing the revision of beliefs when the assumptions underlying them are invalidated [19].

3.1 KLM Framework

The KLM framework, proposed by Kraus, Lehmann, and Magidor extends propositional logic, through the notion of a preferential consequence relation, with a defeasible implication operator ($\mid\sim$) [14] . This allows for the expression of conditional assertions of the form $\alpha \mid\sim \beta$, where α and β are propositional formulas [17]. $\alpha \mid\sim \beta$ reads "typically, if α then β". This preferential structure orders valuations according to their degree of normality, so that reasoning is based on the most typical situations rather than on all situations equally.

For example, the statement $c \mid\sim p$ is equivalent to "cars typically use petrol" and $e \mid\sim \neg p$ expresses that "electric cars typically do not use petrol". Unlike classical implication, these two statements do not contradict one another, the second is simply a more specific default. In this context, defeasible means retractable and any conclusion drawn defeasibly may be withdrawn upon learning contradictory information. This contrasts classical entailment $\models$, where conclusions are permanent and cannot be revised.

3.2 Ranked Interpretations

KLM defeasible implications are interpreted using ranked models that prioritize more typical situations, allowing rules to hold generally while still accommodating exceptions.[16]. A ranked interpretation $\mathcal{R}$ is a function $\mathcal{R} : \mathcal{U} \rightarrow \mathbb{N} \cup \{\infty\}$ that assigns each valuation or world $u \in \mathcal{U}$ a rank. This function must satisfy two conditions. First, at least one valuation must be assigned rank 0, ensuring that there is always a most "typical" world,written $\mathcal{R}(u) = 0$. Second, the ranking must not contain any gaps. This means that if a valuation is assigned rank i, then there must also be valuations at every lower rank j, where $0 \leq j < i$. Formally, for every $i \in \mathbb{N}$, if $\mathcal{R}(v) = i$, then for each $j < i$, there exists some valuation $u \in \mathcal{U}$ such that $\mathcal{R}(u) = j$ [5]. The rank $\mathcal{R}(v)$ represents the degree of typicality of a valuation v. Lower ranks correspond to more typical valuations, while a rank of ∞ indicates that the valuation is impossible [5]. Strict implications, such as $e \rightarrow c$ (classical implication), must be classically held in every possible valuation. Any valuation that violates a strict statement is assigned rank ∞.

Returning to the car example, consider $\mathcal{K} = \{e \rightarrow c, c \mid\sim p, e \mid\sim \neg p\}$. A ranked interpretation satisfying $\mathcal{K}$ would assign rank 0 to ordinary cars that use petrol, rank 1 to electric cars that do not use petrol, and rank ∞ to impossible valuations such as electric cars that use petrol. This ensures that $c \mid\sim p$ and $e \mid\sim \neg p$ coexist without contradiction. This can be seen in the following table:

3.3 Defeasible Entailment

Unlike classical entailment, which requires a conclusion to hold in all possible worlds, defeasible entailment focuses only on the most

Rank	Valuation	Meaning
R_0	$c \wedge p \wedge \neg e$	Ordinary car that uses petrol
R_0	$\neg c \wedge \neg p \wedge \neg e$	Not a car, no petrol, not electric
R_0	$\neg c \wedge p \wedge \neg e$	Not a car but uses petrol
R_0	$\neg c \wedge \neg p \wedge e$	Electric object but not a car
R_1	$c \wedge \neg p \wedge e$	Electric car that does not use petrol
R_1	$c \wedge \neg p \wedge \neg e$	Non-electric car that does not use petrol
R_∞	$c \wedge p \wedge e$	Electric car that uses petrol (impossible)
R_∞	$\neg c \wedge p \wedge e$	Electric object, uses petrol and not a car (impossible)

Table 2: A ranked interpretation satisfying $\mathcal{K}$, showing all possible worlds

typical ones [16].In a ranked interpretation, the most typical valuations are those assigned the lowest rank. A defeasible implication $\alpha \mid\sim \beta$ is satisfied when every minimal α-valuation, that is, every valuation satisfying α at the lowest possible rank, also satisfies β.

A ranked interpretation that satisfies every statement in a knowledge base $\mathcal{K}$ is called a ranked model of $\mathcal{K}$ [20]. $\mathcal{K}$ defeasibly entails $\alpha \mid\sim \beta$, written $\mathcal{K} \mid\approx \alpha \mid\sim \beta$, when $\alpha \mid\sim \beta$ holds in every ranked model of $\mathcal{K}$.

Consider the knowledge base of our example $\mathcal{K}$. To check whether $\mathcal{K}$ defeasibly entails $e \mid\sim \neg p$, we look at the most typical valuations where e holds and that is at rank 1 as $\neg p$ (the antecedent) is true in that valuation, $\mathcal{K} \mid\approx e \mid\sim \neg p$ holds.

Given a knowledge base $\mathcal{K}$, there are many ranked interpretations that satisfy it. Minimal ranked entailment treats every valuation as typical unless the knowledge base forces otherwise [5]. One ranked interpretation is preferred over another if it assigns a lower or equal rank to every valuation. Formally, $\mathcal{R}_1^{\mathcal{K}} \leq_{\mathcal{K}} \mathcal{R}_2^{\mathcal{K}}$ if and only if $\mathcal{R}_1^{\mathcal{K}}(v) \leq \mathcal{R}_2^{\mathcal{K}}(v)$ for every valuation v. The minimal ranked model $\mathcal{R}_{\mathcal{K}}^{RC}$ assigns the lowest possible rank to every valuation while satisfying all statements in $\mathcal{K}$, making this reasoning non-monotonic.

Table 3 shows the minimal ranked model for $\mathcal{K}$. The ordinary car valuation sits at rank 0 because nothing in $\mathcal{K}$ forces it to be exceptional. No valid ranked model of $\mathcal{K}$ can assign lower ranks than these without violating a statement in $\mathcal{K}$.

Rank	Valuation	Meaning
0	$c \wedge p \wedge \neg e$	Ordinary car using petrol
1	$c \wedge \neg p \wedge e$	Electric car not using petrol
∞	$c \wedge p \wedge e$	Electric car using petrol (impossible)

Table 3: Minimal ranked interpretation $\mathcal{R}_{\mathcal{K}}^{RC}$ for $\mathcal{K}$

3.4 Rational Closure[16]

Rational closure is the most fundamental form of defeasible entailment built on the KLM framework [13, 14].It provides a systematic way to determine what follows from a knowledge base $\mathcal{K}$ by ranking statements based on how exceptional they are. More typical information is preferred over more exceptional information [5].Rational closure can be defined semantically, using the minimal ranked model or syntactically, using base ranks and both definitions produce identical inferences [5, 10]. A statement is exceptional with respect to $\mathcal{K}$ if it is false in all the most typical worlds. The base rank $br_{\mathcal{K}}(\alpha)$ of a formula α is the lowest rank at which statement α is no longer exceptional [5]. More general statements get lower ranks and more specific statements get higher ranks. Classical implications, such as $e \rightarrow c$, are not defeasible and must hold in all worlds (rank ∞).

The BaseRank algorithm computes these ranks by iteratively identifying exceptional statements. It takes a defeasible knowledge base $\mathcal{K}$ as input, converts it to its materialisation $\overrightarrow{\mathcal{K}}$, and partitions it into an ordered tuple $(R_0, \ldots, R_{n-1}, R_\infty, n)$. At each iteration, a materialized implication $\alpha \rightarrow \beta$ is deemed exceptional if the current set of statements classically entails $\neg\alpha$, meaning that assuming α is true leads to a contradiction. Only the exceptional statements are carried forward to the next iteration; the non-exceptional ones are collected into R_i and assigned that rank. The process repeats until no further change occurs, at which point any remaining statements are assigned to R_∞ as classical implications [5, 7].

Algorithm 1: BaseRank

Input: A defeasible knowledge base $\mathcal{K}$
Output: An ordered tuple $(R_0, \ldots, R_{n-1}, R_\infty, n)$

```
i := 0;
E_0 := →K;
repeat
    E_{i+1} := {α → β ∈ E_i | E_i ⊨ ¬α};
    R_i := E_i \ E_{i+1};
    i := i + 1;
until E_{i-1} = E_i;
R_∞ := E_{i-1};
if E_{i-1} = ∅ then n := i − 1;
else n := i;
return (R_0, ..., R_{n-1}, R_∞, n);
```

Consider the following knowledge base, which extends the car example with additional defaults:

$$\mathcal{K} = \{e \rightarrow c,\ c \mid\sim p,\ e \mid\sim \neg p,\ c \mid\sim m,\ e \mid\sim b\} \tag{1}$$

This states that electric cars are cars, cars typically use petrol and have motors, while electric cars typically do not use petrol and use batteries. Applying the BaseRank algorithm to $\mathcal{K}$, we first form the materialization $\overrightarrow{\mathcal{K}} = \{e \rightarrow c,\ c \rightarrow p,\ e \rightarrow \neg p,\ c \rightarrow m,\ e \rightarrow b\}$ and trace the following iterations:

- Iteration 0. Starting with $E_0 = \overrightarrow{\mathcal{K}}$, we check which antecedents are exceptional. $E_0 \models \neg e$ because $e \rightarrow c$ and $c \rightarrow p$ together with $e \rightarrow \neg p$ produce a contradiction when e is assumed true. So $e \rightarrow \neg p$, $e \rightarrow b$, and the classical $e \rightarrow c$ all move forward to E_1. The statements $c \rightarrow p$ and $c \rightarrow m$ are not exceptional (assuming c causes no contradiction), so $R_0 = \{c \rightarrow p,\ c \rightarrow m\}$.
- Iteration 1. $E_1 = \{e \rightarrow c,\ e \rightarrow \neg p,\ e \rightarrow b\}$. Without $c \rightarrow p$, the earlier contradiction no longer holds, so $E_1 \not\models \neg e$. The defeasible statements $e \rightarrow \neg p$ and $e \rightarrow b$ are no longer

exceptional, giving $R_1 = \{e \rightarrow \neg p,\ e \rightarrow b\}$. The classical implication $e \rightarrow c$ remains.
- Fixpoint $E_2 = \{e \rightarrow c\}$ and the loop terminates since no antecedents remain exceptional. $R_\infty = \{e \rightarrow c\}$.

This produces the following partition, where every step is a classical entailment check and no defeasible reasoning is required by the algorithm itself [7]:

Rank	Statement
R_∞	$e \rightarrow c$
R_1	$e \mid\sim \neg p,\ \ e \mid\sim b$
R_0	$c \mid\sim p,\ \ c \mid\sim m$

Table 4: Base ranks for the extended $\mathcal{K}$

To evaluate the query $\mathcal{K} \mid\approx_{RC} e \mid\sim \neg p$, We check whether e is exceptional

Using all ranks, the materialized knowledge $c \rightarrow p$ and $e \rightarrow c$ together contradict $e \rightarrow \neg p$, making e exceptional and R_0 is removed. With only R_1 and R_∞ remaining, e is no longer exceptional. We then check whether the remaining statements entail $e \rightarrow \neg p$, which they do. Therefore, $\mathcal{K} \mid\approx_{RC} e \mid\sim \neg p$ holds.

Formally, a defeasible implication $\alpha \mid\sim \beta$ is in the rational closure of $\mathcal{K}$ if $br_{\mathcal{K}}(\alpha) < br_{\mathcal{K}}(\alpha \wedge \neg\beta)$ or $br_{\mathcal{K}}(\alpha) = \infty$ [16]. That is, $\alpha \mid\sim \beta$ holds when α is more typical than the situation where α holds but β does not.The Rational Closure algorithm computes this by repeatedly removing the lowest rank until α is no longer exceptional, then checking whether the remaining statements classically entail $\alpha \rightarrow \beta$ [7]. Giordano et al. [10] showed that the ranks produced by BaseRank correspond directly to the ranks of valuations in the minimal ranked model, confirming that rational closure reduces defeasible entailment to a series of classical entailment checks [5].

This rank-removal strategy has a known limitation: when e is exceptional, the entire rank $R_0 = \{c \mid\sim p,\ c \mid\sim m\}$ is discarded, even though only $c \mid\sim p$ was involved in the conflict. As a result, the query $e \mid\sim m$ (electric cars typically have motors) cannot be concluded by Rational Closure, since $c \mid\sim m$ was discarded along with the offending statement. This is the drowning problem, which motivates the lexicographic approach described below.

3.5 Lexicographic Closure

Rational closure is too cautious by design, that is when an antecedent conflicts with a knowledge base, the algorithm discards the entire rank containing the conflict. If a statement falls under an atypical subclass, no typical properties of the superclass can be inherited [6]. Lexicographic closure addresses this by formalizing a more human, presumptive reasoning[15]. Rather discarding an entire rank, it identifies and removes only the specific statements causing the conflict and therefore preserving useful information. Lehmann defines lexicographic closure using four key principles that guide how defaults are applied [16][15].

- **Presumption of Typicality** states that typical outcomes are assumed unless there is evidence against them. For example, if $\mathcal{K} = \{c \mid\sim p\}$, then from c we assume p holds in typical cases.
- **Presumption of Independence** states that unrelated properties do not affect each other. If $\mathcal{K} = \{c \mid\sim p,\ c \mid\sim \neg r\}$, we still accept $c \wedge r \mid\sim p$ because there is no rule linking r to p. The truth of r does not change what we know about p
- **Priority to Typicality** resolves conflicts between typicality and independence. When these lead to different conclusions, the result based on typical cases is preferred. For example, if $\mathcal{K} = \{c \mid\sim p,\ c \wedge e \mid\sim \neg p\}$, then for $c \wedge e \wedge r$, we conclude $\neg p$. The more typical case for electric cars overrides the general rule
- **Respect for Specificity** states that more specific information takes priority over general information. If two rules conflict, the rule with the stronger condition is chosen. For instance, $c \wedge e$ is more specific than c, so conclusions based on $c \wedge e$ override those based only on c

Lexicographic closure resolves conflicts between defaults by preferring conclusions that violate fewer and less serious rules [15]. A seriousness ordering is introduced over subsets of defeasible implications. The idea is that not all violations are equally important. A valuation is preferred if it violates fewer rules, and if it avoids violating more specific rules[12, 15]. Specificity is determined by base rank, where more specific rules have higher rank.

A basis is the most preferred set of defaults that can still be kept without contradicting α. A subset $B \subseteq \mathcal{K}$ is a basis for α if the materialization $\overrightarrow{B}$ is consistent with α, and B is maximal under the seriousness ordering [15].

Lexicographic closure is therefore defined as follows. A defeasible implication $\alpha \mid\sim \beta$ follows from $\mathcal{K}$ if β holds in every basis of α. Formally, $\mathcal{K} \mid\approx_{LC} \alpha \mid\sim \beta$ if and only if $\overrightarrow{B} \cup \{\alpha\} \models \beta$ for every basis B [15].

Algorithm 2: LexC – Lexicographic Closure

Input: A defeasible knowledge base $\mathcal{K}$ and a defeasible query $\alpha \mid\sim \beta$
Output: true if $\mathcal{K} \mid\approx_{LC} \alpha \mid\sim \beta$, false otherwise

$(R_0, \ldots, R_{n-1}, R_\infty, n) :=$ BaseRank($\mathcal{K}$);
$R := \bigcup_{j=0}^{n} R_j$;
$i := 0$; $m := 0$;
while $\overrightarrow{R_\infty} \cup \overrightarrow{R} \models \neg\alpha$ ***and*** $R \neq \emptyset$ **do**
  $R := R \setminus R_i$;
  $m := |R_i| - 1$;
  $R_{i,m} := \bigvee_{X \in \text{Subsets}(R_i,m)} \bigwedge_{x \in X} x$;
  while $\overrightarrow{R_\infty} \cup \overrightarrow{R} \cup \{\overrightarrow{R_{i,m}}\} \models \neg\alpha$ ***and*** $m > 0$ **do**
    $m := m - 1$;
    $R_{i,m} := \bigvee_{X \in \text{Subsets}(R_i,m)} \bigwedge_{x \in X} x$;
  $R := R \cup \{R_{i,m}\}$;
  $i := i + 1$;
return $\overrightarrow{R_\infty} \cup \overrightarrow{R} \models \alpha \rightarrow \beta$;

The difference from rational closure appears in the inner loop of the algorithm. Instead of removing an entire rank R_i when α is exceptional, lexicographic closure keeps as many statements from that rank as possible.

It does this by constructing a formula $R_{i,m}$ that represents all subsets of R_i of size m. This formula ensures that at least one group of m statements from R_i holds. If keeping m statements still makes α exceptional, the algorithm reduces m and tries again. This continues until it finds the largest subset of R_i that does not cause a conflict.

The remaining statements are then added back to R, and the algorithm moves to the next rank. This means that only the statements responsible for the conflict are removed, while all others are preserved [5, 15].

To illustrate, consider our example from earlier $\mathcal{K} = \{e \to c, c \mathrel{|\!\sim} p, c \mathrel{|\!\sim} m, e \mathrel{|\!\sim} \neg p\}$ and the query $e \mathrel{|\!\sim} m$, which Rational Closure cannot conclude. Using the BaseRank output $R_0 = \{c \to p,\ c \to m\}$, $R_1 = \{e \to \neg p\}$, $R_\infty = \{e \to c\}$, we trace LexC step by step:

(1) Initialize: $R = R_0 \cup R_1 = \{c \to p,\ c \to m,\ e \to \neg p\}$, $i = 0$.
(2) Outer loop check: $\overrightarrow{R_\infty} \cup \overrightarrow{R} \models \neg e$? Yes: $e \to c$ and $c \to p$ together with $e \to \neg p$ produce a contradiction. So the loop body executes.
(3) Rank 0: Remove R_0 from R, leaving $R = \{e \to \neg p\}$. Set $m = |R_0| - 1 = 1$. Compute:
$$R_{0,1} = (c \to p) \vee (c \to m)$$
This says: at least one of the two rank-0 defaults holds.
(4) Inner loop check: Does $\overrightarrow{R_\infty} \cup \overrightarrow{R} \cup \{R_{0,1}\} \models \neg e$? No. The disjunction allows a model where only $c \to m$ holds and not $c \to p$, removing the contradiction. So the inner loop does not execute and m stays at 1.
(5) Add back statement: $R := \{e \to \neg p,\ (c \to p) \vee (c \to m)\}$. Increment $i = 1$.
(6) Outer loop check again: $\overrightarrow{R_\infty} \cup \overrightarrow{R} \models \neg e$? No. The loop exits.
(7) Return: Does $\overrightarrow{R_\infty} \cup \overrightarrow{R} \models e \to m$? Yes: $e \to c$ combined with the disjunction $(c \to p) \vee (c \to m)$ ensures that in any model where e holds and c holds, m is reachable via $c \to m$. LexC returns **true**.

The motor default $c \mathrel{|\!\sim} m$ was preserved because it played no role in the conflict over petrol, which is exactly the presumptive behavior that distinguishes lexicographic closure from rational closure [5, 15]

3.6 Relevant Closure

Rational closure is the most cautious approach to defeasible inference, by drawing only the most prototypical conclusions while lexicographic closure is more presumptive. Relevant closure, first proposed by Casini et al. [4], lies in the middle of the two in that it respects where rational closure is cautious while still drawing additional inferences based on relevance [5, 12]. A limitation of rational closure is that when it encounters a conflict at a given rank, it discards the entire rank, including statements that had nothing to do with the conflict. Relevant closure addresses this by identifying only the statements directly responsible for the conflict and restricting removal to those statements [4]. Given a knowledge base $\mathcal{K}$ and a query $\alpha \mathrel{|\!\sim} \beta$, relevant closure only identifies a subset $R \subseteq \mathcal{K}$ of defeasible statements that are relevant to the query [4]. Given a knowledge base $\mathcal{K}$ and a query $\alpha \mathrel{|\!\sim} \beta$, relevant closure identifies a subset $R \subseteq \mathcal{K}$ of defeasible statements that are relevant to the query [4], where relevance is based on justification. The relevant set R is built by identifying the α-justifications, which are the smallest subsets of $\mathcal{K}$ whose materialization classically entails $\neg\alpha$ [4, 12], the statements directly responsible for making α exceptional.

The knowledge base is then partitioned into $\langle R, \overline{R}\rangle$, where $\overline{R} = \mathcal{K} \setminus R$ contains all irrelevant statements. The algorithm proceeds like rational closure, but only statements in R can be removed. Statements in $\overline{R}$ are never removed [4].

There are two variants. Basic relevant closure considers all statements across all justifications as relevant and Minimal relevant closure, which is more selective. It takes only the lowest-ranked statements from each justification as relevant. [12**?**]. This prevents the relevant set from becoming too large, as it does in lexicographic closure.

Algorithm 3: RelevantClosure

Input: A defeasible knowledge base $\mathcal{K}$, a defeasible query $\alpha \mathrel{|\!\sim} \beta$, and a partition $\langle R, \overline{R}\rangle$ of $\mathcal{K}$
Output: true if $\mathcal{K} \mid\!\approx_{RC} \alpha \mathrel{|\!\sim} \beta$, false otherwise

```
(R_0, ..., R_{n-1}, R_∞, n) := BaseRank(K);
i := 0;
R' := R;
while R_∞^→ ∪ R̄^→ ∪ R'^→ ⊨ ¬α and R' ≠ ∅ do
    R' := R' \ {R_i ∩ R};
    i := i + 1;
return R_∞^→ ∪ R̄^→ ∪ R'^→ ⊨ α → β;
```

To illustrate minimal relevant closure, consider $\mathcal{K} = \{p \mathrel{|\!\sim} b, b \mathrel{|\!\sim} f, p \mathrel{|\!\sim} \neg f, b \mathrel{|\!\sim} w\}$ and the query $p \mathrel{|\!\sim} w$ (penguins typically have wings). The base rank assigns rank 0 to $b \mathrel{|\!\sim} f$ and $b \mathrel{|\!\sim} w$, and rank 1 to $p \mathrel{|\!\sim} b$ and $p \mathrel{|\!\sim} \neg f$.

The justification for why p is exceptional is $\{p \mathrel{|\!\sim} b, b \mathrel{|\!\sim} f, p \mathrel{|\!\sim} \neg f\}$. The irrelevant statement $b \mathrel{|\!\sim} w$ is not part of this justification. The relevant set R contains only the lowest-ranked statement in the justification, namely $b \mathrel{|\!\sim} f$ and the irrelevant statements, including $b \mathrel{|\!\sim} w$, go into $\overline{R}$ and are never removed.

When the algorithm runs, it removes $b \mathrel{|\!\sim} f$ from R' at rank 0. After removal, p is no longer exceptional. The final check asks whether $\overrightarrow{R_\infty} \cup \overrightarrow{\overline{R}} \cup \overrightarrow{R'} \models p \to w$. Since $b \mathrel{|\!\sim} w$ was never removed, the answer is yes. Relevant closure concludes $p \mathrel{|\!\sim} w$, which rational closure cannot.

Relevant closure does not satisfy all KLM properties and is therefore not LM-rational [5]. Despite this, it draws more conclusions than rational closure in cases where specific statements are irrelevant to the exception, making it practically useful even if it is theoretically weaker than lexicographic closure.

4 Pedagogical Tools for Defeasible reasoning

A survey of existing tools reveals two distinct categories: pedagogical tools for designed to support learning in classical logic, and defeasible reasoners which have no pedagogical foundations. Despite the growing importance of defeasible reasoning in AI, no pedagogical tool for defeasible reasoning was found. This is a significant gap, as students encountering defeasible reasoning for the first time may need structured support for understanding the reasoning steps behind the conclusions produced by defeasible inferences.

4.1 Pedagogical Classical Reasoning tools

Although no pedagogical tool for defeasible reasoning was found, examining existing tools for classical logic is valuable, as they demonstrate the effect of logic education in practice, and highlight the gap that remains for defeasible reasoning. Iltis, developed at TU Dortmund by Geck et al. [9], is a web-based interactive tool that is designed to teach logic across the full reasoning workflow, covering modeling, transformation, and reasoning. The system supports propositional, modal, and first-order logic through a uniform interface which provides structured feedback, tailored to each exercise type. Iltis has been used in second-year computer science courses, and had a positive impact on students. Although Iltis covers classical logic, it provides no support for nonmonotonic or defeasible reasoning. A student working through Iltis exercises would gain a solid foundation in classical inference, but would find no guidance in working with defeasible concepts such as how conclusions can be revised, how defaults are ordered by exceptionality, or how defeasible algorithms such as Lexicographic Closure could resolve conflicts by preserving relevant defaults. Iltis sits entirely on the classical side of the gap that motivates the development of an educational tool for defeasible reasoning.

4.2 Defeasible Reasoner

Beyond pedagogical tools for classical logic, there are systems that implement defeasible reasoning and these are primarily designed for research rather than educational purposes. These systems illustrate practical implementations of defeasible reasoning. The Tweety Project[21] is a collection of Java libraries that offer a general interface for research in knowledge representation. It serves as a framework for implementing various approaches to logical artificial intelligence and implements defeasible logic programming via DeLP, which differs from the KLM preferential reasoning framework used in this project. It lacks built-in rational or lexicographic closure and the system targets researchers with strong programming skills as interaction is through Java APIs without graphical user interactive interface. Tweety offers no pedagogical support, such as guided exercises, feedback mechanisms, or instructional scaffolding, making it a research tool rather than a learning tool. RaMP [17] is a plugin for the Protege ontology editor, implementing prototypical and presumptive reasoning over ontologies, enabling defeasible inferencing over defeasible information in the ontology. It uses axiom annotations in Protege to represent defeasible information, enabling users to rank axioms. RaMP can visualize these rankings, a feature relevant to the tool being developed in this project. It operates in Protege using description logic instead of propositional logic and is aimed at research, not education. SILK [11] is a semantic rule language known for its expressive nature and use of courteous prioritized defeasible reasoning. It features a graphical user interface, facilitates knowledge input, answers queries, and allows for justification browsing of defeasible information as an Eclipse IDE plugin. This plugin supports exploration by automatically tracking query results in response to user modifications of rules. Justifications are presented in a tree format, which allows for in-depth exploration of the knowledge to comprehend the query. Unlike KLM preferential reasoning, SILK employs manual ranking, which sets it apart from the approach used in this project. Nonetheless, the features of justification browsing and query exploration are directly applicable to the tool being developed.

4.3 Gaps

The tools examined reveal a clear gap. On one side, pedagogical tools such as Iltis provides a structured and feedback driven learning environment, that is effective for learning classical logic but offers no support for defeasible reasoning. On the other side, systems such as Tweety, RaMP, and SILK demonstrated that defeasible reasoning can be implemented into a capable system, but none of them have pedagogical support infrastructure. No existing tool bridges this gap. A student seeking to understand defeasible reasoning concepts such as the Lexicographic Closure algorithm, or why certain statements are preserved while others are discarded, has no dedicated tool to support that understanding. This absence motivates the development of this pedagogical tool, which aims to make the reasoning steps of Lexicographic Closure transparent and accessible to students encountering defeasible reasoning for the first time.

5 Conclusion

This review investigated the theoretical and practical foundations essential for developing a pedagogical tool centered on defeasible reasoning. It began by examining propositional logic and the limitations of classical monotonic reasoning, emphasizing the necessity for defeasible reasoning. The KLM framework was introduced as the formal basis, offering semantic backing for defeasible conditionals through ranked interpretations and defeasible entailment. Three defeasible entailment methods were analyzed. Rational closure, the most conservative approach, simplifies defeasible entailment to a series of classical entailment checks by assigning base ranks to statements and eliminating the lowest ranks when conflicts occur. Its main drawback is entire ranks being discarded even if only one statement causes a conflict. Lexicographic closure addresses this by retaining as many statements from a conflicting rank as possible, preserving defaults unrelated to the exception. Relevant closure takes a different approach by pinpointing the specific statements causing the exceptionality and limiting removal to those statements alone. A review of existing tools revealed the absence of a dedicated pedagogical tool for defeasible reasoning. While some tools support classical logic, they do not extend to defeasible reasoning. Research tools such as Tweety, RaMP, and SILK implement defeasible reasoning but are designed for expert users. This gap underscores the need for a new tool that makes the reasoning process behind lexicographic closure visible and accessible to students and researchers interested in defeasible reasoning.

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